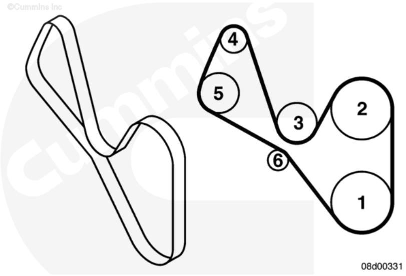
Trainings

General Information



Due to the number of cooling fan drive belt arrangements, this procedure does not cover all available cooling fan drive belt routings.

To ensure the cooling fan drive belt is routed correctly upon installation, make a diagram of the cooling fan belt routing prior to removing the belt as shown in the illustration.

The cooling fan belt routing typically consists of the following components, but may not include all of them:

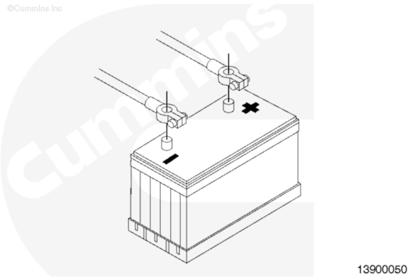
1. Crankshaft Pulley/Vibration Damper
2. Fan Pulley
3. Water Pump Pulley
4. Alternator Pulley
5. Refrigerant Compressor Pulley
6. Belt Tensioner Pulley
7. Idler Pulley(s) (**not** shown)

Note : Some engine driven belts are installed/supplied by the vehicle's OEM. See the OEM's service manual on removal and installation procedures.

Preparatory Steps

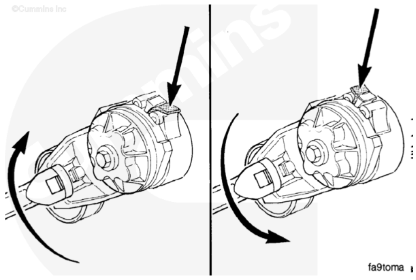
⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



- Disconnect the batteries.

Remove



⚠ CAUTION ⚠

The belt tensioner is spring-loaded and must be pivoted away from the drive belt. Pivoting in the wrong direction can result in damage to the belt tensioner.

The belt tensioner winds in the direction that the spring tang is bent over the tensioner body. To loosen the tension on the belt, rotate the tensioner to wind the spring tighter.

⚠ CAUTION ⚠

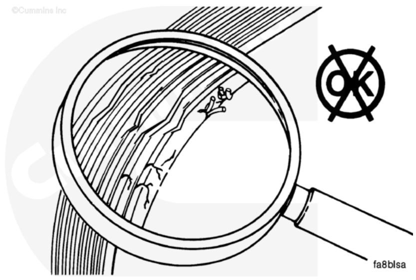
Applying excessive force in the opposite direction of windup or after the tensioner has been wound up to the positive stop can cause the tensioner arm to crack or break.

Note : Make a diagram of the belt arrangement prior to removing the drive belt to aid in installation and ensure proper routing of the cooling fan drive belt.

Pivot the tensioner in the direction of the spring tang to remove the drive belt.

Remove the drive belt.

Inspect for Reuse



Inspect the belt for:

- Cracks
- Glazing
- Tears or cuts
- Excessive wear.

Replace the belt if any damage is found.

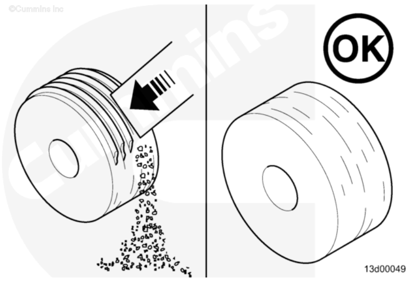
Inspect the idler and drive pulleys for wear or cracks.

Plastic pulleys often have a build-up of road dirt and belt material that is **not** to be confused with wear.

The dirt can be removed with a suitable tool to check for wear.

Clean, check and reuse idlers with a build up of dirt, rather than replacing.

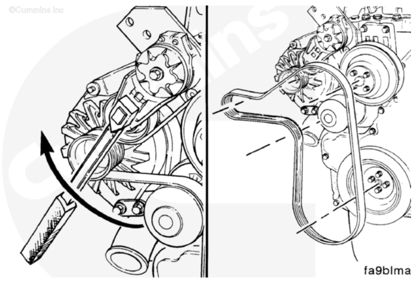
Inspect the belt tensioner. Refer to Procedure 008-087 (/qs3/pubsys2/xml/en/procedures/40/40-008-087-tr.html).



Install

⚠ CAUTION ⚠

The belt tensioner is spring-loaded and must be pivoted away from the drive belt. Pivoting in the wrong direction can result in damage to the belt tensioner.



Route the drive belt on the engine using the belt diagram created in the remove step. Do **not** install the belt over the water pump pulley at this time.

Pivot the tensioner in the direction of the spring tang and install the drive belt, slipping the belt over the water pump pulley last.

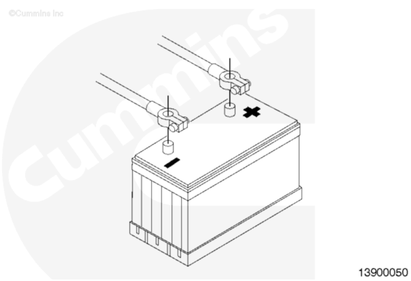
Release the tensioner to apply tension to the drive belt.

Check the alignment of the belt with the tensioner and the rest of the front end auxiliary drive.

Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



- Connect the batteries.

- Operate the engine and check for belt squeal. Excessive belt squeal indicates belt slippage.
- If belt squeal is present, check the routing of the belt and make sure that the belt is installed correctly on each pulley.

Last Modified: 31-Mar-2006